Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. In the continuous-time state-space model form: 1 / 1 point

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t)$$

$$z(t) = Cx(t) + Du(t) + v(t),$$

what are the names of the two equations?

- ○ The first equation is the "dynamics equation" and the second equation is the "output equation."
- ○ The first equation is the "input equation" and the second equation is the "output equation."
- ○ The first equation is the "measurement equation" and the second equation is the "process equation."
- ◉ The first equation is the "state equation" and the second equation is the "output equation."

Correct
Yes. This is correct.

2. In the continuous-time state-space model form: 1 / 1 point

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t)$$

$$z(t) = Cx(t) + Du(t) + v(t),$$

which signal or signals is/are measured inputs to the model? (Select all that apply.)

- [ ] $x(t)$.
- [x] $u(t)$.

Correct
Yes. This is the measured input to the model.

- [ ] $w(t)$.
- [ ] $z(t)$.
- [ ] $v(t)$.

3. In the continuous-time state-space model form: 1 / 1 point

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t)$$

$$z(t) = Cx(t) + Du(t) + v(t),$$

what are the names of the matrices A, B, C, and D?

- ○ A is the "evolution matrix," B is the "input matrix," C is the "combination matrix," and D is the "instantaneous matrix."
- ○ A is the "system matrix," B is the "feedthrough matrix," C is the "output matrix," and D is the "input matrix."
- ○ A is the "dynamics matrix," B is the "process matrix," C is the "sensor matrix," and D is the "feedforward matrix."
- ◉ A is the "system matrix," B is the "input matrix," C is the "output matrix," and D is the "feedthrough matrix."

Correct
Yes. This is correct.